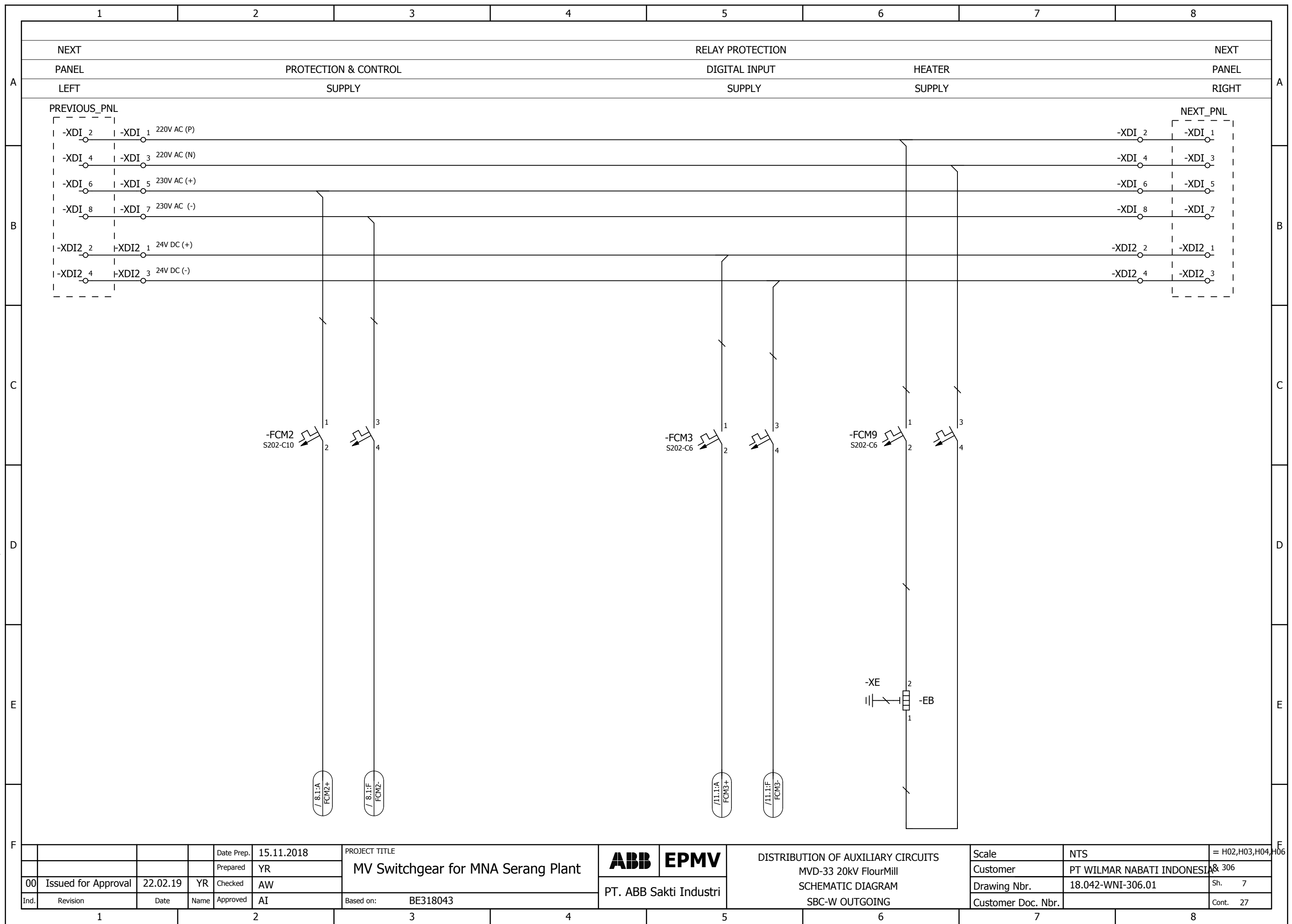


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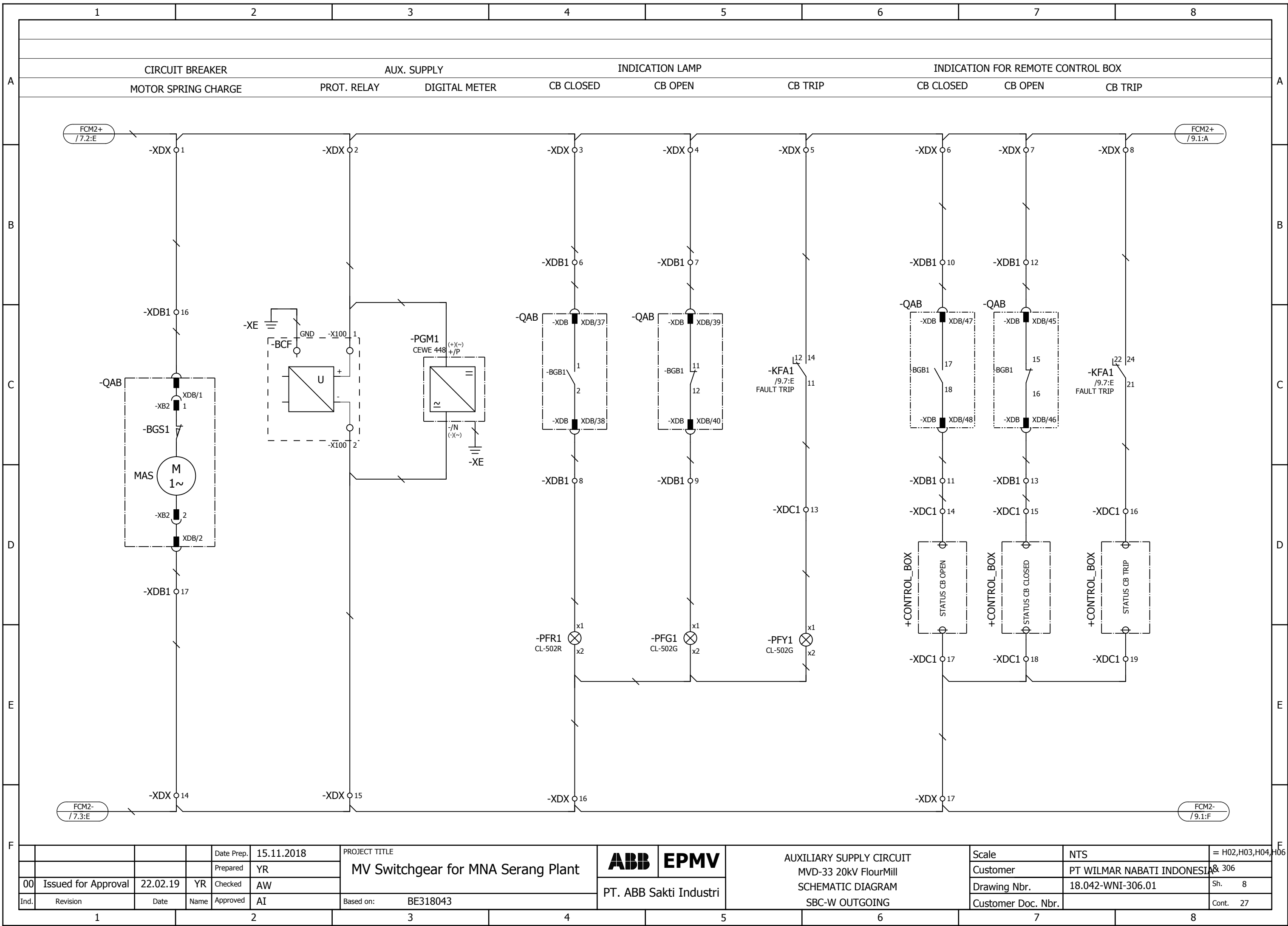
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F															
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING H02,H03,H04,H06															
				Date Prep. 15.11.2018		PROJECT TITLE				COVER SHEET		Scale		NTS	
				Prepared YR		MV Switchgear for MNA Serang Plant				MVD-33 20kV FlourMill		Customer		PT WILMAR NABATI INDONESIA	
00		Issued for Approval		22.02.19 YR		Checked AW		PT. ABB Sakti Industri				Drawing Nbr.		18.042-WNI-306.01	
Ind.		Revision		Date		Name		Approved AI		Based on: BE318043				Customer Doc. Nbr.	
1		2		3		4		5		6		7		8	

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	1	2	3	4	5	6	7	8					
A	SHEET INDEX								A				
	PAGE			PAGE NAME				REVISION					
	==MVD-33=H02,H03,H04,H06&306/1			COVER SHEET				00					
	==MVD-33=H02,H03,H04,H06&306/2			Index of Sheet				00					
	==MVD-33=H02,H03,H04,H06&306/3			REFERENCE DESIGNATIONS				00					
	==MVD-33=H02,H03,H04,H06&306/4			REFERENCE DESIGNATIONS				00					
	==MVD-33=H02,H03,H04,H06&306/5			REFERENCE DESIGNATIONS				00					
	==MVD-33=H02,H03,H04,H06&306/6			REFERENCE DESIGNATIONS				00					
	==MVD-33=H02,H03,H04,H06&306/7			DISTRIBUTION OF AUXILIARY CIRCUITS				00					
	==MVD-33=H02,H03,H04,H06&306/8			AUXILIARY SUPPLY CIRCUIT				00					
		==MVD-33=H02,H03,H04,H06&306/9			CIRCUIT BREAKER CONTROL				00				
==MVD-33=H02,H03,H04,H06&306/10			AUXILIARY MOTOR & RELAY				00						
==MVD-33=H02,H03,H04,H06&306/11			DIGITAL INPUT/OUPUT RELAY				00						
==MVD-33=H02,H03,H04,H06&306/12			SPARE CONTACT				00						
==MVD-33=H02,H03,H04,H06&306/13			CURRENT TRANSFOMER				00						
==MVD-33=H02,H03,H04,H06&306/14			VOLTAGE TRANSFORMER				00						
		==MVD-33=H02,H03,H04,H06&306/15			LV DOOR LAYOUT				00				
	==MVD-33=H02,H03,H04,H06&306/16			LV BOX LAYOUT				00					
	==MVD-33=H02,H03,H04,H06&306/17			PART LIST				00					
	==MVD-33=H02,H03,H04,H06&306/18			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDA									
	==MVD-33=H02,H03,H04,H06&306/19			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDB1									
	==MVD-33=H02,H03,H04,H06&306/20			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDC1									
	==MVD-33=H02,H03,H04,H06&306/21			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDC2									
	==MVD-33=H02,H03,H04,H06&306/22			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDI									
	==MVD-33=H02,H03,H04,H06&306/23			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDI2									
	==MVD-33=H02,H03,H04,H06&306/24			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDI3									
	==MVD-33=H02,H03,H04,H06&306/25			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDI6									
	==MVD-33=H02,H03,H04,H06&306/26			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDV									
	==MVD-33=H02,H03,H04,H06&306/27			Terminal diagram ==MVD-33=H02,H03,H04,H06+-XDX									
										E			
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	Index of Sheet MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03,H04,H06
				Prepared	YR	MV Switchgear for MNA Serang Plant		Customer			PT WILMAR NABATI INDONESIA	& 306	
00	Issued for Approval	22.02.19	YR	Checked	AW			Drawing Nbr.			18.042-WNI-306.01	Sh. 2	
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 27
	1	2	3	4	5	6	7	8					

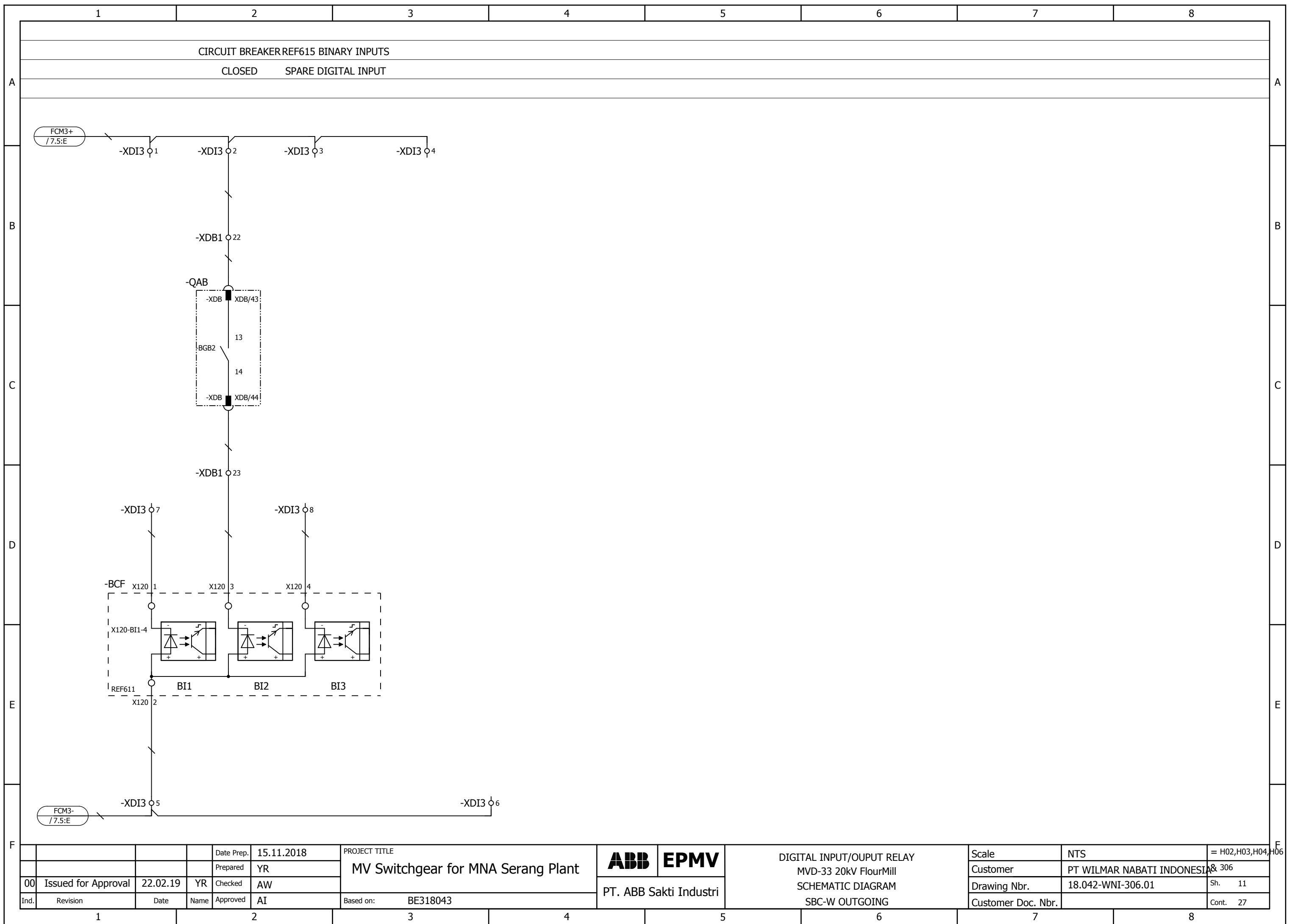


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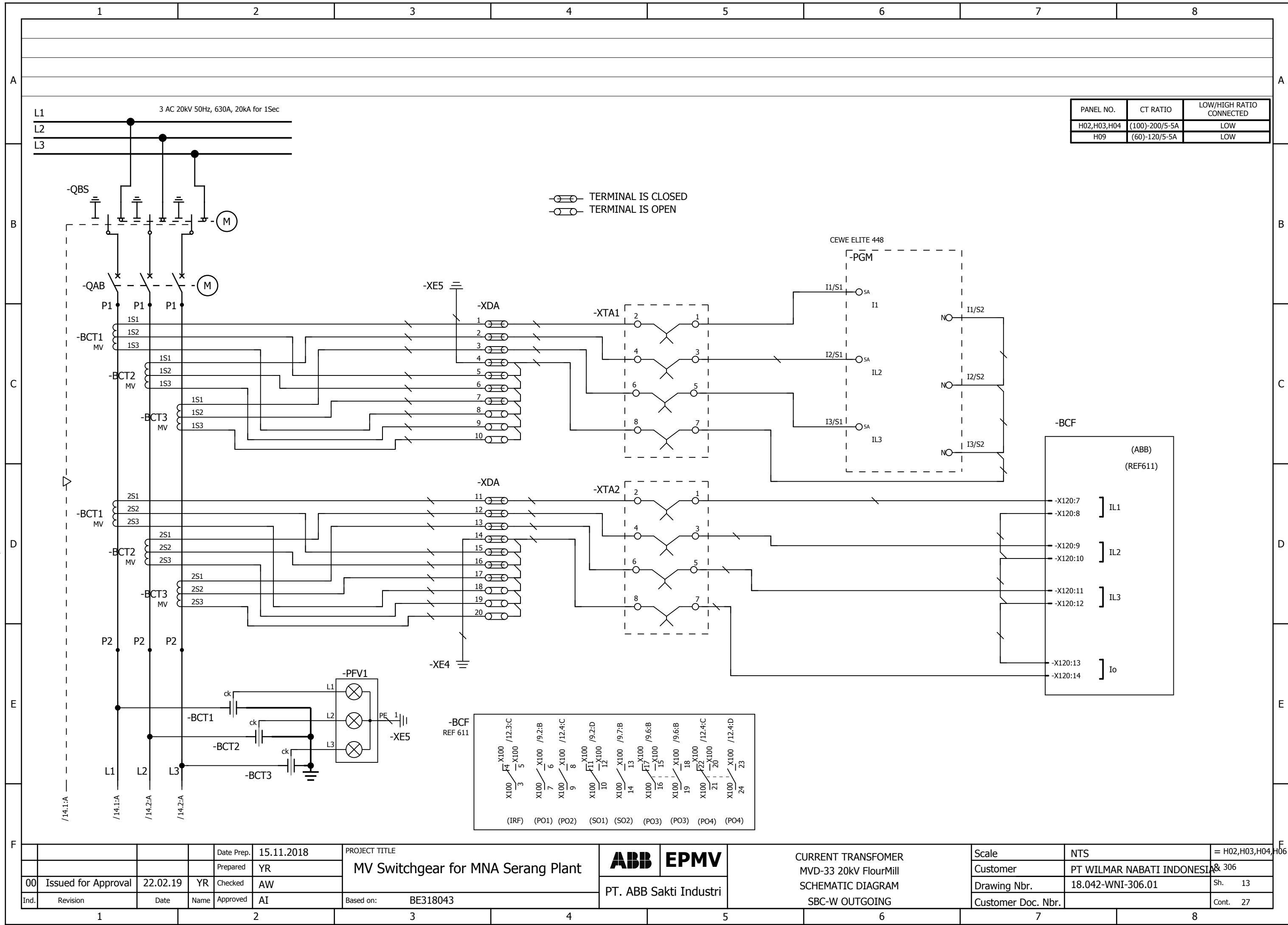


00		Issued for Approval	22.02.19	YR	Checked	AW	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB EPMV	AUXILIARY SUPPLY CIRCUIT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03,H04,H06
Ind.		Revision	Date	Name	Approved	AI							
							Based on:	BE318043	PT. ABB Sakti Industri				
											Customer	PT WILMAR NABATI INDONESIA	& 306
											Drawing Nbr.	18.042-WNI-306.01	Sh. 8
											Customer Doc. Nbr.		Cont. 27

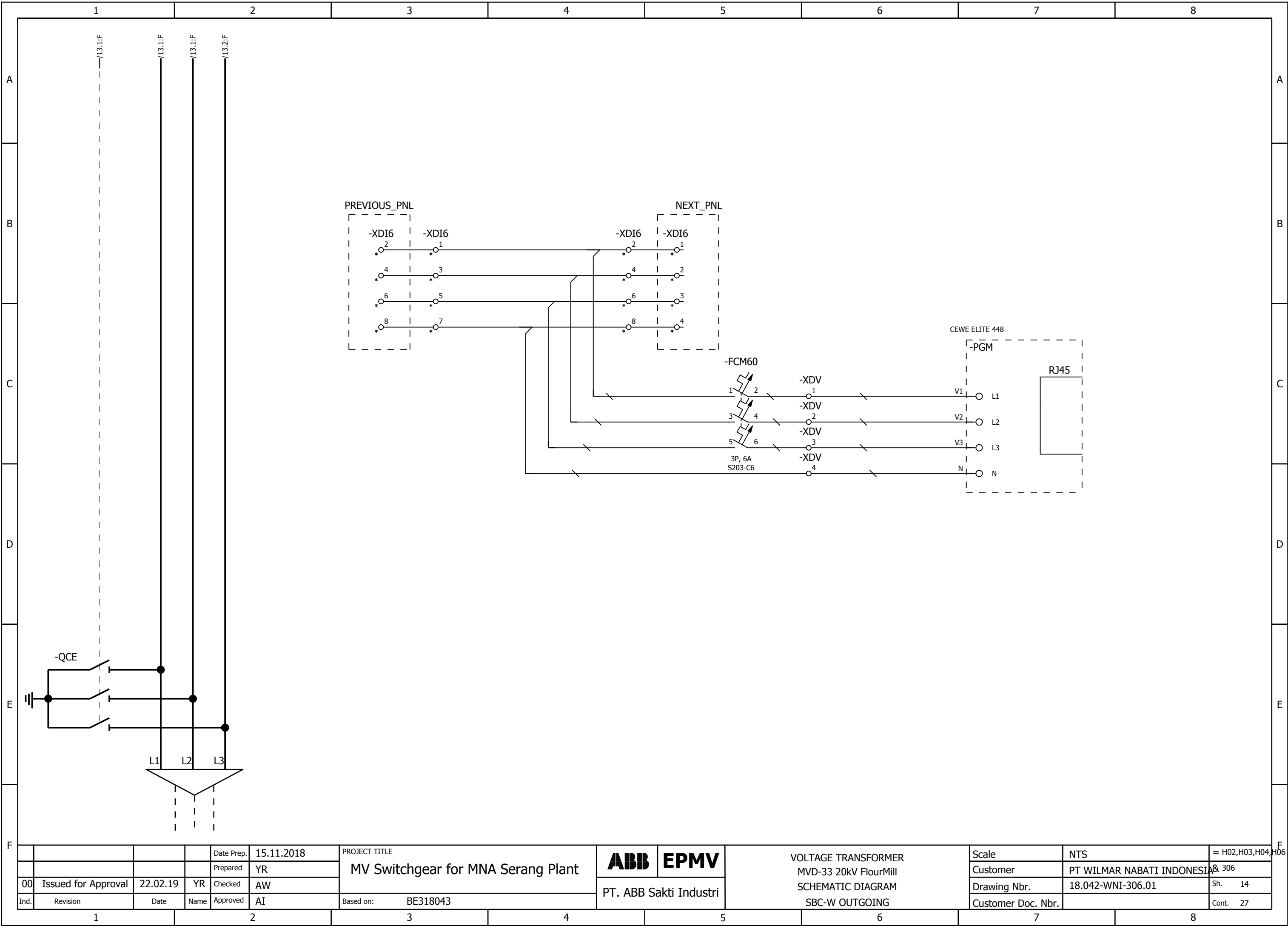
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			AUXILIARY MOTOR & RELAY MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03,H04,H06
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 306
00	Issued for Approval	22.02.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-WNI-306.01	Sh. 10	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043		Customer Doc. Nbr.			Cont. 27	



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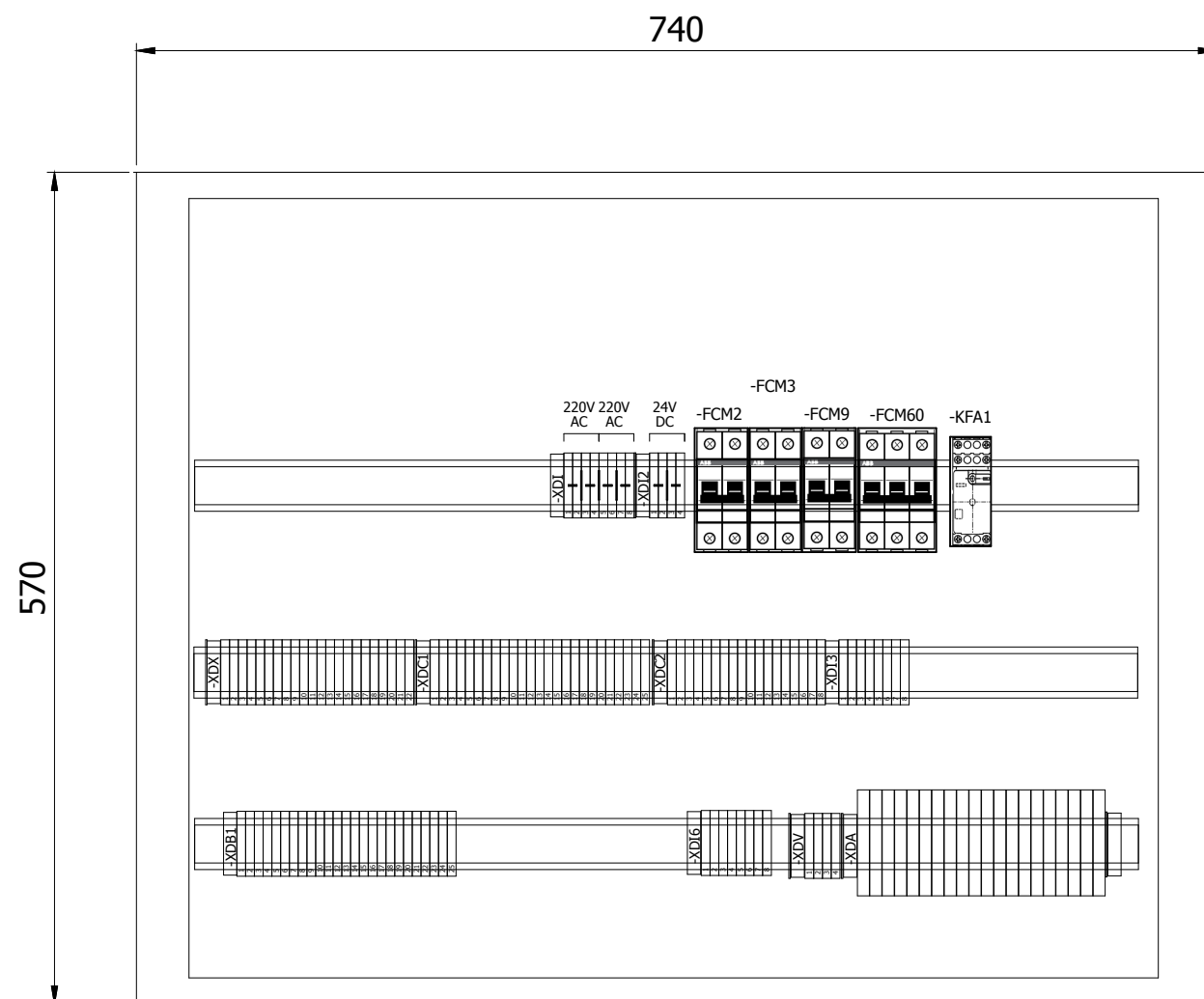


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00	Issued for Approval	22.02.19	YR	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB EPMV	VOLTAGE TRANSFORMER MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03,H04,H06
				Prepared	YR						Customer	PT WILMAR NABATI INDONESIA
				Checked	AW						Drawing Nbr.	18.042-WNI-306.01
				Approved	AI						Customer Doc. Nbr.	Sh. 14
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PT. ABB Sakti Industri	SBC-W OUTGOING			Cont. 27

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03,H04,H06
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 306
00	Issued for Approval	22.02.19	YR	Checked	AW					Drawing Nbr.	18.042-WNI-306.01	Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 27	

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1	2		3		4		5		6		7		8				
Summarized parts list														IDABB_F02_005-IDABB			
														Carry over: 0.00			
A	TAG ID				DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY	REMARKS		A	
	-BCF				Feeder Management Relay REF611HBAABC1NN11G				ABB		REF611HBAABC1NN11G		1	/8.2:C			
	-EB				HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1	/7.6:E			
B	-FCM2				MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1	/7.2:C			
	-FCM3;-FCM9				MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2	/7.5:C;/7.6:C		B	
	-FCM60				MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1	/14.5:C			
	-KFA1				Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1	/9.7:E			
C	-PFG1				PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1	/8.5:E			
	-PFR1				PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1	/8.4:E		C	
	-PFY1				PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1	/8.6:E			
	-PGM1				POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1	/8.3:C			
D	-SFC				PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1	/9.2:B			
	-SFO				PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1	/9.4:B		D	
	-SFS				LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1	/9.2:B			
	-XTA1;-XTA2				Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2	/13.4:C;/13.4:D			
E																	
F																	
				Date Prep.	15.11.2018		PROJECT TITLE			ABB	EPMV	PART LIST			Scale	NTS	= H02,H03,H04,H06
				Prepared	YR		MV Switchgear for MNA Serang Plant					MVD-33 20kV FlourMill			Customer	PT WILMAR NABATI INDONESIA	& 306
	00	Issued for Approval	22.02.19	YR	Checked	AW				PT. ABB Sakti Industri		SCHEMATIC DIAGRAM			Drawing Nbr.	18.042-WNI-306.01	Sh. 17
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043					SBC-W OUTGOING			Customer Doc. Nbr.		Cont. 27
	1		2		3		4		5		6		7		8		

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F				E		D		C		B		A	
</													

F										E	D	C	B	A					
==MVD-33=H02,H03,H04,H06+-XDC1										REMARK	CABLE	EQUIPMENT	WIRE NO.	TYPE	Terminal number	Wire No	EQUIPMENT	CABLE	REMARK
BLACK 1,5 mm²																			BLACK 1,5 mm²
BLACK 1,5 mm²																			BLACK 1,5 mm²
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BLACK 1,5 mm²																			

[illegible]

[illegible]

F	E	D	C	B	A	REMARK	CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK	1	2	3	4	5	6	7	8
																REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.	A	B	C	D	E	F	
Ind.	Revision	Date	Name	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	Terminal diagram	Scale		= H02,H03,H04,H06										
				Prepared	YR						Customer	PT WILMAR NABATI INDONESIA & 306											
				Checked	AW						Drawing Nbr.			Sh.	23								
				Approved	AI						Based on:	BE318043		Customer Doc. Nbr.		Cont.	27						
1	2	3	4	5	6	7	8	A	B	C	D	E	F										

[illegible]

F		E		D		C		B		A	
==MVD-33=H02,H03,H04,H06+-XDI6		REMARK	CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK
		BLACK 1,5 mm ²		-PREVIOUS_PNL-XDI6:2		1 ●	PT 2.5				
				-FCM60:1		2 ●	PT 2.5			-NEXT_PNL-XDI6:1	
				-PREVIOUS_PNL-XDI6:4		3 ●	PT 2.5				
		BLACK 1,5 mm ²		-FCM60:3		4 ●	PT 2.5			-NEXT_PNL-XDI6:2	
				-PREVIOUS_PNL-XDI6:6		5 ●	PT 2.5				
		BLACK 1,5 mm ²		-FCM60:5		6 ●	PT 2.5			-NEXT_PNL-XDI6:3	
				-PREVIOUS_PNL-XDI6:8		7 ●	PT 2.5				
		BLACK 1,5 mm ²		-XDV:4		8 ●	PT 2.5			-NEXT_PNL-XDI6:4	
		REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS.									
		△ OUTGOING CONTROL CABLE (Supply by Customer).									
		● BRIDGE BAR.									
		✕ PARTITION PLATE.									
		(TEST JUMPER.									

Ind.		Revision		Date		Name		Approved		Date Prep.		15.11.2018		PROJECT TITLE		MV Switchgear for MNA Serang Plant		ABB EPMV		Terminal diagram		Scale				= H02,H03,H04,H06	
										Prepared		YR										Customer		PT WILMAR NABATI INDONESIA		Sh. 25	
										Checked		AW						PT. ABB Sakti Industri				Drawing Nbr.				Cont. 27	
										Based on:		BE318043										Customer Doc. Nbr.					

F	E	D	C	B	A	1	2	3	4	5	6	7	8	F			
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT WILMAR NABATI INDONESIA	= H02,H03,H04,H06
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV</						

